False Sharing and Scalability in Multicore Processors: Impact of Cache-Line Data Layout on Parallel Performance

Walter Alfaro, Allan Arrieta

***Abstract*—Modern multicore processors keep private caches coherent at the granularity of a cache line, typically 64 bytes, rather than individual variables. When threads running on different cores write to independent variables that happen to reside in the same cache line, the coherence protocol forces the line to migrate repeatedly between cores, even though no data is logically shared. This phenomenon, known as false sharing, generates invalidation traffic and cache-to-cache transfers that can severely limit the scalability of parallel programs and is a recurring source of performance regressions in large systems such as the Linux kernel. This paper investigates how the distribution of variables across cache lines affects the performance of a parallel program on a hybrid x86-64 Intel Core i7-1260P (Alder Lake) processor, which combines performance and efficiency cores. We study a microbenchmark in which each thread, pinned to a dedicated core, updates its own independent counter, comparing a layout where the counters are stored contiguously against layouts in which they are separated through alignment and padding, as well as per-thread accumulation. The analysis relates the observed execution time and scalability with the number of threads to the behavior of the MESI-family coherence protocol, and uses hardware performance counters and the Linux `perf c2c` tool to identify contended cache lines through HITM (load hit in a modified line) events. In addition, the memory and cache-capacity overhead introduced by padding is quantified, together with the mitigation strategies documented by the Linux kernel community, in order to characterize the trade-off between coherence-traffic reduction and memory footprint and to derive practical guidelines for cache-aware data layout.**

***Index Terms*—False sharing, cache coherence, MESI, multicore processors, cache line, padding, performance analysis, perf c2c, Linux kernel.**

I. Introduction

PARALLEL programs are expected to run faster as more cores are added. A well-known obstacle to this expectation is *false sharing*: threads that never touch each other's data can still slow each other down when their data happen to share a cache line [1], [2]. Because the effect depends on memory layout rather than on the algorithm, it is invisible in the source code and is usually discovered only through profiling [3], [4].

[TODO: 1 parrafo: por que importa hoy (kernel, servidores, laptops hibridas). Citar la doc del kernel como fuente del problema, no como prueba de magnitud.]

Most of the guidance on false sharing comes from a small group of sources: processor vendors, the developers of the detection tools, and research prototypes evaluated by their own authors. Rather than taking that guidance at face value, this paper treats each recommendation as a claim to be tested. We pose three questions:

1) **RQ1:** How much does placing independent counters in the same cache line degrade performance and scalability on a contemporary hybrid laptop processor?
2) **RQ2:** Which commonly cited claims about false sharing (magnitude of the slowdown, need for 128-byte padding, reliability of HITM-based detection) hold under our model and measurements, and which depend on the context in which their sources were produced?
3) **RQ3:** What does mitigation cost in memory footprint, and when is that cost worth paying?

[TODO: Contribuciones: (1) simulador MESI propio y abierto; (2) mediciones en i7-1260P con perf c2c; (3) contraste simulacion vs. hardware; (4) lectura critica de las fuentes. Ultimo parrafo: organizacion del paper.]

II. Background

A. Coherence at Cache-Line Granularity

Private caches in a multicore processor are kept coherent by a hardware protocol. The unit tracked by the protocol is the cache line (64 bytes on current x86-64 processors [5]), not the individual variable. Consequently, two variables that reside in the same line are treated by the hardware as a single shared object [6], [7].

B. The MESI Protocol

[TODO: Describir los 4 estados (M, E, S, I), las transacciones BusRd, BusRdX y BusUpgr, y una tabla o diagrama de transiciones. Fuentes: Hennessy–Patterson y Nagarajan et al. Aclarar que Intel usa una variante (MESIF) y que el detalle exacto no es publico.]

C. True Sharing vs. False Sharing

[TODO: Definicion formal (Torrellas et al.): un fallo es de false sharing si la palabra que se invalida no es la que se usa. Condiciones: varios nucleos, misma linea, al menos una escritura [2]. Figura: 4 contadores de 8 B en una linea de 64 B.]

D. Detecting False Sharing with `perf c2c`

[TODO: Que es un HITM (load que encuentra la linea en estado M en otra cache), como perf c2c lo muestrea y que reporta (lineas, offsets, nodos/CPUs) [3], [8]. Anotar desde ya la limitacion: es muestreo y se basa en loads.]

TABLE I
CRITICAL READING OF THE MAIN SOURCES

Source	Context / possible bias	How we test it
Kernel docs [2]	Written by an Intel engineer; cases from large servers	Magnitude on a 4-P-core laptop
Mario [3]	Co-author of the tool he documents	Detection under few or store-only accesses
Intel manual [5]	Vendor document; claims rarely quantified	64 B vs. 128 B padding
Drepper [9]	2007 hardware	Whether costs still hold
Torrellas [1]	Simulated 1990s bus machines	Line-size sweep in our simulator
SHERIFF [4]	Authors evaluate their own tool	**[TODO: revisar trabajos posteriores]**

III. RELATED WORK AND CRITICAL REVIEW OF SOURCES

A. What the Sources Claim

[TODO: Resumen breve de cada fuente: Torrellas 1994 (false sharing vs. localidad espacial y tamano de linea), Liu y Berger 2011 (SHERIFF), doc del kernel (patrones y mitigaciones), Mario 2016 (perf c2c), manual de Intel (recomendaciones de layout y prefetcher), Drepper 2007 (costo de coherencia).]

B. Who Wrote Them, and Under What Conditions

[TODO: Para cada fuente, contexto y posible sesgo. Verificar cada punto contra la fuente antes de afirmarlo.]

IV. METHODOLOGY

A. Workload

Each of N threads increments its own 64-bit counter I times. No counter is ever read or written by another thread, so any coherence traffic between threads is, by construction, false sharing. Four data layouts are compared:

- **packed:** counters stored contiguously (8 B apart), so up to eight counters share one 64-byte line;
- **padded:** counters separated by a stride s (64 or 128 B), so each occupies its own line;
- **local:** each thread accumulates in a private variable and writes the shared array once at the end;
- **shared:** all threads increment the *same* counter (true sharing), used as a reference.

B. Trace-Driven MESI Simulator

We implemented a MESI coherence simulator in C (available in the project repository, `sim/mesi_sim.c`). Its model is deliberately simple so that every result can be traced to a protocol event:

- N cores, each with a private set-associative LRU cache;
- a single, serialized snooping bus carrying BusRd, BusRdX and BusUpgr transactions;
- an atomic increment modeled as a read-for-ownership (BusRdX) followed by a write, and, alternatively, as a separate load and store;

TABLE II
SIMULATOR LATENCY PARAMETERS (CYCLES AT 4.7 GHZ)

Event	Cycles	Origin
Private cache hit	4	assumed
Locked RMW, line already owned	18	calibrated
Data from a modified line (HITM)	47	calibrated
Data from LLC / clean copy	40	assumed
Main memory	200	assumed
Upgrade (S→M)	30	assumed

- a per-core clock; at each step the core with the lowest clock issues its next access, so contention emerges from timing instead of from a fixed interleaving;
- an optional *adjacent-line prefetch* that, on a miss, also fetches the other line of the aligned 128-byte pair.

The simulator counts hits, misses, bus transactions, invalidations, cache-to-cache transfers from a modified line (the analogue of a HITM), write-backs, and the memory occupied by the counters. Latencies are model parameters (Table II); a sensitivity sweep over the HITM latency is also reported.

The two parameters that dominate the workload were calibrated from our own measurements on the P-cores (Section V): the cost of an uncontended locked increment (padded layout, one thread, 3.87 ns) and the cost of one line transfer, taken from the slope of the packed layout with four threads (about 10 ns per transfer). The remaining values are rough estimates; they barely affect this workload because, after warm-up, it never misses to the LLC or to memory. Note that calibrating with data from the same platform means that agreement at those two points is not an independent validation; only the predictions for other thread counts and layouts are.

[TODO: Aclarar que el bus serializado es una simplificacion: el i7-1260P usa un ring y una L3 compartida. Explicar por que aun asi el modelo predice el crecimiento lineal (la linea disputada es una sola: solo un nucleo puede tenerla en M a la vez).]

Four sweeps are run (`sim/sweep.sh`): (i) scaling with N for each layout; (ii) line size versus counter stride; (iii) adjacent-line prefetch with 64-byte and 128-byte padding; and (iv) sensitivity to the HITM latency.

C. Hardware Experiment

The experiment runs on an Intel Core i7-1260P (Alder Lake) with four performance cores (P-cores, two hardware threads each, private 1.25 MiB L2, up to 4.7 GHz), eight efficiency cores (E-cores, up to 3.4 GHz, organized in two clusters of four that share a 2 MiB L2) and an 18 MiB L3 shared by all cores, under Linux 7.0 with the `performance` frequency governor. The benchmark (`benchmarks/fs_bench.c`, compiled with GCC 13.3 at `-O2`) implements the four layouts of the simulator, with either a locked increment (`__atomic_fetch_add`, which compiles to `lock add`) or a plain `volatile` increment. Each thread is pinned to one core, and three groups of cores are compared: one thread per physical P-core (CPUs 0, 2, 4, 6), four E-cores of one cluster (8–11), and E-cores split across both clusters (8, 12, 9, 13). Each configuration performs 5×10^7 increments per thread

TABLE III
SIMULATED NS PER INCREMENT, 4 THREADS (PACKED COUNTERS AT DISTANCE d)

	Distance between counters d (bytes)					
Line	8	16	32	64	128	256
32 B	40.0	40.0	4.1	4.1	4.1	4.1
64 B	40.0	40.0	40.0	4.1	4.1	4.1
128 B	40.0	40.0	40.0	40.0	4.1	4.1
256 B	40.0	40.0	40.0	40.0	40.0	4.1

TABLE IV
HITM SAMPLES REPORTED BY `perf c2c`, 4 THREADS

Layout	**P-cores**	**E same**	**E cross**
packed, atomic	18 631	54	293
shared, atomic	12 951	186	97
padded, atomic	0	0	1
local	0	0	0
packed, volatile	0	0	20
Measured slowdown, packed vs. padded (atomic)			
	8.4×	9.2×	11.6×

and is repeated ten times after one warm-up run. Contended lines are recorded with `perf c2c record` for each layout and core group.

V. RESULTS

A. Simulation

Table III shows the simulated cost of an increment with four threads as a function of the cache-line size and the distance between counters. The result is binary: whenever two counters fall in the same line, every increment pays a line transfer; as soon as the distance reaches the line size, the cost drops to that of an uncontended locked increment. The boundary moves with the line size, which reproduces in our model the argument of Torrellas et al. [1] that longer lines increase false sharing.

With the adjacent-line prefetch enabled and 64-byte padding, the simulator records only one to four prefetch-induced HITM events per run (for 2 to 8 threads), all during warm-up. Once each core owns its line in the M state it never misses again, so the prefetcher is never triggered. **[TODO: Mencionar el barrido de sensibilidad (`hitm_sensitivity.csv`): el costo de packed escala linealmente con la latencia HITM supuesta.]**

B. Measurements on the i7-1260P

Fig. 1 shows the cost per increment for each layout and core group with locked increments. With one thread all layouts except local cost the same (3.87 ns on P-cores, 5.32 ns on E-cores; about 18 cycles in both cases). With two or more threads, the packed and shared layouts grow roughly linearly with the thread count, while padded remains almost flat. With four threads, packed is $8.4\times$ slower than padded on P-cores, $9.2\times$ on E-cores sharing an L2, and $11.6\times$ on E-cores in different clusters (Fig. 2). Padding to 128 bytes was within 4% of 64-byte padding in every configuration and is omitted from the figure. All configurations were repeated ten times; the coefficient of variation was below 2% except for the local layout, whose runs last only a few milliseconds.

The type of increment changes the picture. With plain `volatile` increments (a load, an add and a store, without `lock`), the packed layout is only $1.15\times$ slower than padded on P-cores, but $3.8\times$ to $5.5\times$ slower on E-cores (Fig. 2). On a P-core a single thread completes about one `volatile` increment per cycle.

Table IV summarizes the `perf c2c` recordings. On P-cores, the tool reports one contended line with 18 631 HITM samples and shows each thread writing at a different offset (0x0, 0x8, 0x10, 0x18), exactly the pattern described by Mario [3]. Padded and local layouts produce no HITM. However, the same packed workload on E-cores yields only 54 and 293 HITM samples, and the `volatile` packed layout yields almost none on any group.

C. Simulation vs. Hardware

After calibrating two parameters per core group, the simulator predicts the packed layout within $\pm 7\%$ for two to four threads in all three groups (Fig. 1). The fitted line-transfer cost is 47 cycles on P-cores, 42 cycles between E-cores sharing an L2, and 50 cycles between E-core clusters. The model fails in two places: it predicts that packed and shared cost the same, whereas on two P-cores packed is 40% slower than shared (21.6 vs. 15.4 ns); and, since it has no store buffer, it cannot reproduce the near absence of false sharing for `volatile` increments on P-cores. **[TODO: Explicar las dos discrepancias en la Discusion.]**

VI. DISCUSSION

A. Is False Sharing as Costly as True Sharing?

[TODO: En el modelo, packed y shared generan el mismo trafico. Comprobar en hardware y discutir que implica: para el hardware no existe diferencia.]

B. Does 128-Byte Padding Matter?

[TODO: En el modelo, el prefetch de la linea adyacente solo provoca un evento aislado cuando los contadores ya estan en estado estable: la recomendacion de 128 B dependeria de que haya fallos repetidos. Contrastar con el manual de Intel y con la medicion.]

C. Generalizing from Servers to a Hybrid Laptop

[TODO: Las fuentes describen servidores con muchos nucleos; aqui hay 4 P-cores y una L3 compartida. Que cambia y que no.]

D. Limits of HITM-Based Detection

[TODO: perf c2c muestrea loads; que pasa con patrones dominados por stores.]

E. The Cost of Mitigation

[TODO: 8 B vs. 64 B vs. 128 B por contador: 8x o 16x mas memoria, mas lineas de cache y presion de TLB. Cuando no vale la pena (contadores poco escritos).]

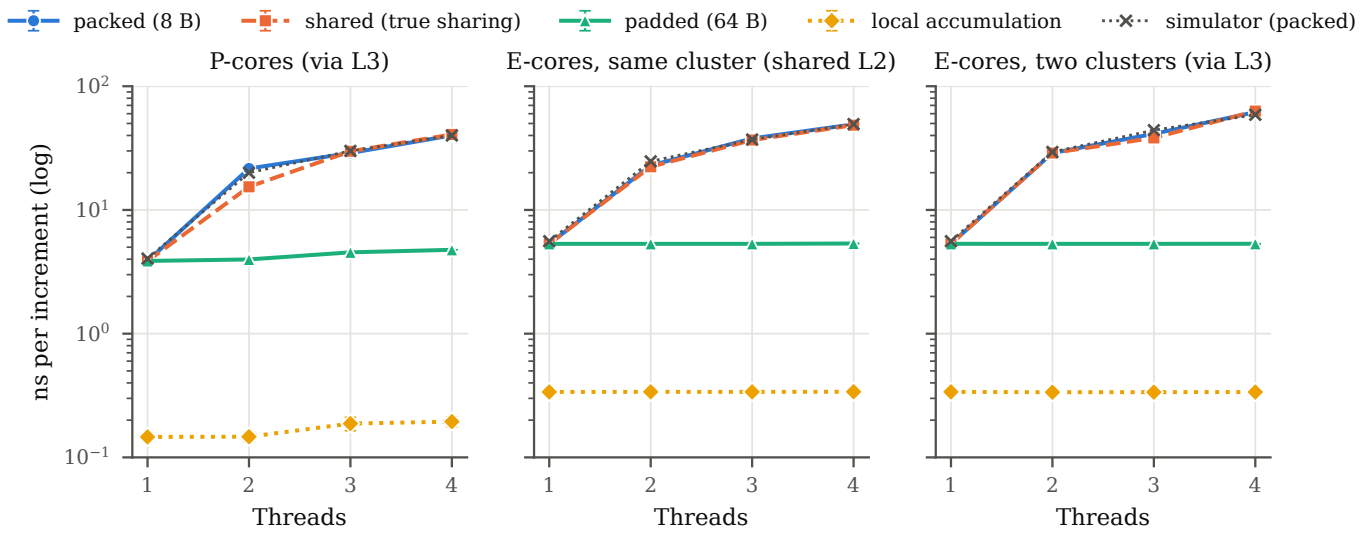


Fig. 1. Cost per locked increment versus thread count on the i7-1260P (mean of 10 runs; error bars show one standard deviation, mostly smaller than the markers). Each panel pins the threads to a different group of cores. The dotted line is the prediction of our MESI simulator for the packed layout, using the line-transfer latency calibrated for each group (Table II).

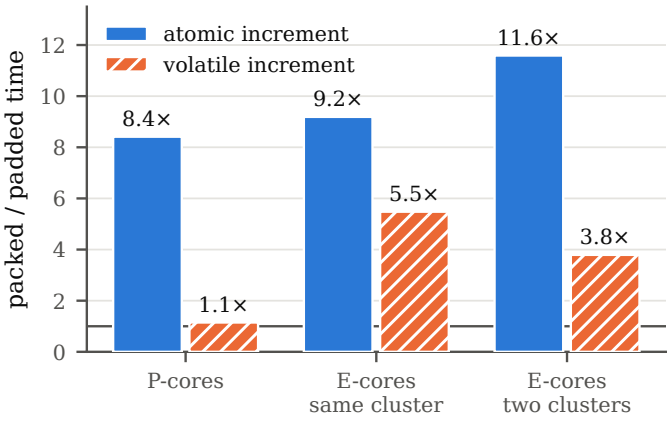


Fig. 2. Slowdown of the packed layout relative to 64-byte padding with four threads, for locked (`lock add`) and plain `volatile` increments.

F. Threats to Validity

[TODO: Simplificaciones del simulador, frecuencia variable/turbo, scheduler, microbenchmark como peor caso.]

VII. Conclusions

[TODO: Responder RQ1–RQ3 en un parrafo cada una y dar guias practicas.]

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